RAG Learning

Required Practicals — PAG1 to PAG8

OCR GCSE Physics A (J249) — Practical Activity Group summaries

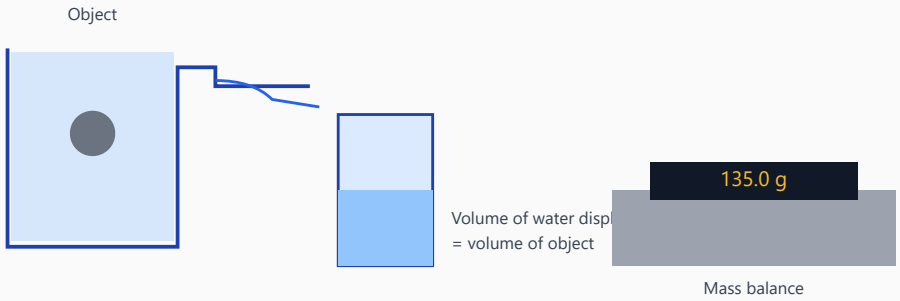
There are eight **Practical Activity Groups (PAGs)** you need to know for OCR A GCSE Physics. You may be asked about apparatus, method, variables, or how to analyse the results in the written papers. Each one is summarised below with an apparatus diagram and key expected results.

PAG1 — Density of Solid Objects and Liquids

Aim: determine the density of regular solids, irregular solids, and a liquid.

Independent variable: object / liquid used.
Dependent variable: density (calculated).
Control variable: temperature (affects density of water noticeably), use same balance.

Measuring density by displacement (irregular solid)



Method: Regular solid: measure mass with a balance; measure dimensions with a ruler / vernier calliper; calculate V. Then ρ = m / V.
Irregular solid: measure mass. Use a eureka can full to the spout — lower the object; collect displaced water in a measuring cylinder. Volume of water = volume of object.

Liquid: place an empty measuring cylinder on a balance; tare to zero; pour in a known volume and record the mass.

⚠️ **Safety:** Wear eye protection. Handle glassware carefully — wet benches cause slipping. Dry any spilled water.

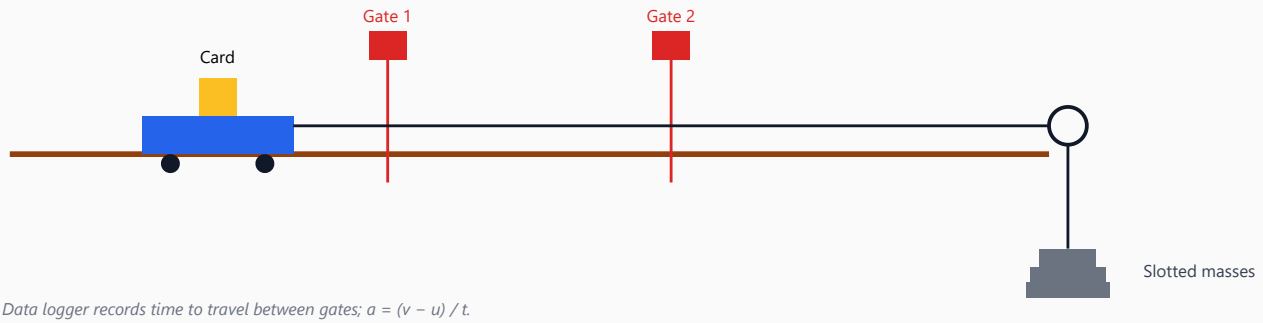
Exam Tip: Common error: reading the volume at the top of the meniscus rather than the bottom. Always read to the bottom of the meniscus at eye level.

PAG2 — Force, Mass and Acceleration (Newton's Second Law)

Aim: investigate how the acceleration of a trolley depends on (a) accelerating force and (b) total mass.

Independent variable: force (via slotted masses on a hanger) *or* total mass of the trolley system.
Dependent variable: acceleration.
Control variables: surface used, same trolley, same light gate spacing.

Trolley, light gates, pulley and mass hanger



Data logger records time to travel between gates; a = (v − u) / t.

Method: 1. Set up: ramp (use a slight tilt only to compensate for friction so trolley moves at steady velocity when given a gentle push).
2. Place one light gate near the top of the ramp and a second light gate a measured

distance away.
3. Attach a card to the trolley so it can interrupt the light gate beams.
4. Add masses to the hanger to accelerate the trolley — record the mass (so F = mg is known).
5. Release the trolley, record the times / velocities from each gate.
6. Calculate acceleration a = (v − u) / t.
7. **Part A**: keep trolley mass constant, vary hanging mass (accelerating force). Plot a against F → straight line through origin.
8. **Part B**: keep F constant, add mass *to the trolley*. Plot a against 1/m → straight line through origin.

Expect: $a \propto F$ (at constant m) and $a \propto 1/m$ (at constant F). Consistent with $F = m\,a$.

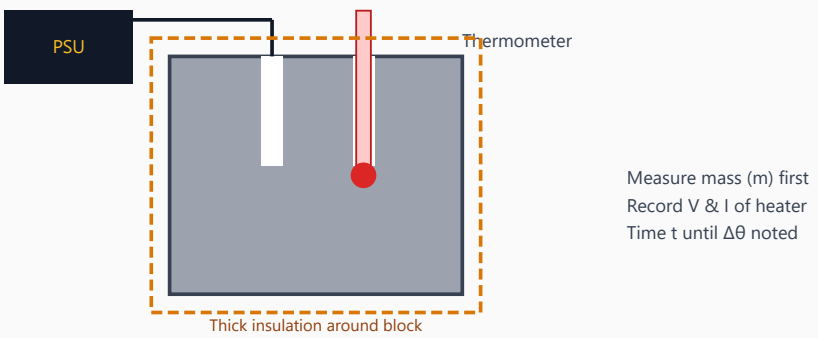
⚠️ **Safety:** Clamp the pulley securely. Place a folded towel / tray below the hanger to catch falling masses.

PAG3 — Specific Heat Capacity

Aim: determine the specific heat capacity (c) of a metal block.

Independent variable: time (as block heats).
Dependent variable: temperature rise of block.
Control variables: block material, voltage / current through heater, same insulation.

Metal block, immersion heater and thermometer



Method: 1. Weigh the block (mass m). Insulate it well (fibreglass / polystyrene).
2. Insert heater in one hole; thermometer in the other with a drop of oil to improve thermal contact.
3. Record initial temperature θ_0.
4. Switch on heater, record V and I. Start timer.
5. Record temperature every 30 s for 10 minutes.
6. Plot temperature vs time. Take $\Delta\theta$.
7. Energy supplied $E = V \times I \times t$.
8. $c = E / (m \times \Delta\theta)$.

Specific heat capacity: $\mathbf{c = V\,I\,t / (m\,\Delta\theta)}$

Exam Tip: The measured c is usually *higher* than the true value because some energy is lost as heat to the surroundings — this is why good insulation is needed.

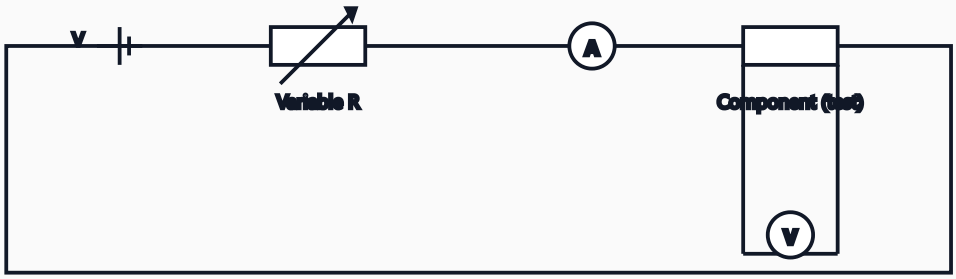
⚠️ **Safety:** The block and heater get very hot — do not touch for several minutes after switch-off. Mains-operated PSUs should be set to the correct voltage.

PAG4 — I-V Characteristics of Components

Aim: investigate how current through a component varies with pd for a fixed resistor, a filament lamp, and a diode.

Independent variable: pd across component.
Dependent variable: current through component.
Control variables: ambient temperature, same component throughout.

Circuit: variable resistor controls pd; ammeter in series, voltmeter in parallel



Method: 1. Build the circuit with the component under test in the main loop, ammeter in series, voltmeter in parallel with the component.
2. Adjust the variable resistor from minimum → maximum pd across the component, recording pairs (V, I) at ~0.5 V steps.
3. **Reverse the cell connections** and repeat to get negative V values (important for the diode).
4. Plot I (y-axis) against V (x-axis).

Component	Shape of I–V graph	Interpretation
Fixed resistor at constant T	Straight line through origin	Ohmic — R is constant
Filament lamp	S-shape through origin, flatter at extremes	Non-ohmic; resistance rises as filament heats
Diode	Zero current in reverse; rapid rise above threshold (~0.6 V) in forward	Only conducts one way above threshold

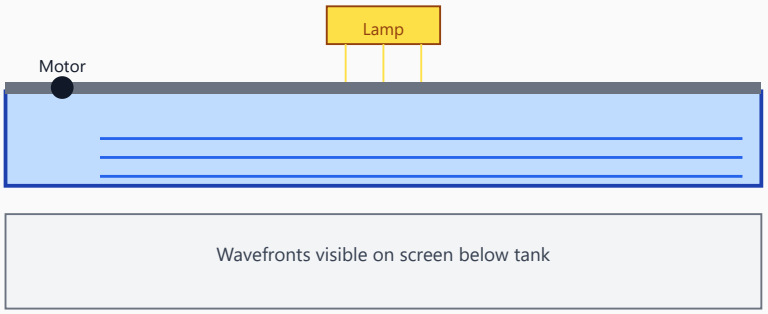
⚠️ **Safety:** The filament lamp gets hot. Keep pd within the component's rating or it will burn out. Disconnect between readings if the lamp glows bright.

PAG5 — Wave Speed (Ripple Tank and Stretched String)

Aim: measure the speed of waves on a water surface and of transverse waves on a stretched string.

Method A — Water waves in a ripple tank

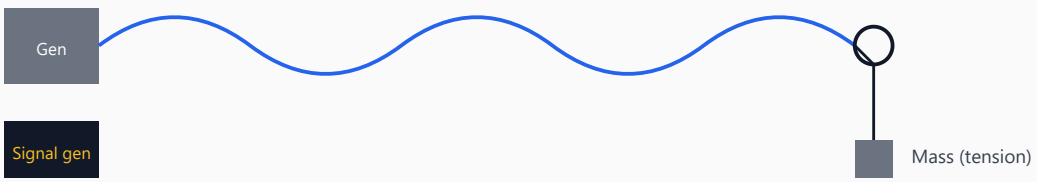
Ripple tank setup



Method: Use a vibrating bar at known frequency f. Light above casts a shadow on a screen below. Measure wavelength λ by timing a known number of waves crossing a fixed distance (or by placing a ruler next to the shadow and measuring the spacing). Wave speed $v = f \times \lambda$.

Method B — Transverse waves on a string

Vibration generator, pulley, masses



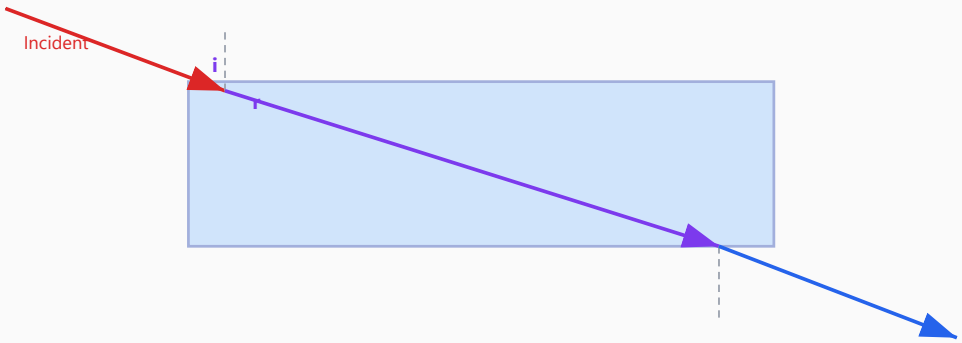
Method: Fix a known length of string between a vibration generator and a pulley with a mass providing tension. Use a signal generator to set the frequency. Increase f until a stable standing wave forms (e.g. two loops). From $\lambda = 2L/n$ (n = number of loops) and given f, compute v.

⚠️ **Safety:** Keep fingers clear of the vibration generator. Wear eye protection if strings might snap.

PAG6 — Refraction (Glass Block)

Aim: investigate how a light ray changes direction when it passes through a rectangular glass block.

Ray box and glass block on plain paper



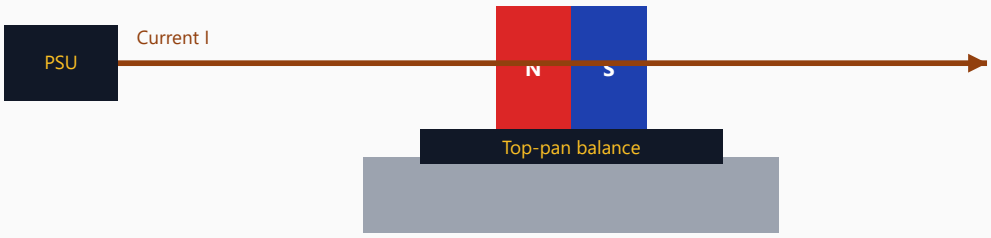
Method: 1. Place the glass block on plain paper; draw around it.
2. Shine a narrow ray from a ray box at an angle of incidence i onto one long side.
3. Mark where the ray enters and leaves the block.
4. Remove the block; join up the dots to reconstruct the path inside the block.
5. Draw normals at the entry and exit points.
6. Measure i and r with a protractor.
7. Repeat for several angles of incidence (e.g. 20°, 30°, 40°, 50°, 60°).
8. Plot r against i (not a straight line) or sin r against sin i (straight through origin — gradient = 1 / refractive index).

Exam Tip: Use a **thin** pencil to mark ray positions. Use at least three dots along each ray so you can draw straight lines accurately. Measure angles *from the normal*, not from the block surface.

PAG7 — Magnetic Field Around a Wire / Current in a Magnetic Field

Aim: investigate the magnetic field pattern around a current-carrying wire, and the force on a current-carrying conductor in a magnetic field.

Current-carrying wire between magnet poles on top-pan balance



Method: 1. Clamp a pair of opposite magnetic poles so the field passes horizontally between them.
2. Place the magnets on a top-pan balance; tare to 0.
3. Pass a wire horizontally between the poles (perpendicular to the field).
4. Switch on the current; note the reading on the balance.
5. A change in reading shows the force on the wire (Newton's 3rd law: wire pushes on the magnets, magnets push wire up/down).
6. Vary the current I; plot mass change against I. Line should be straight through origin — $F \propto I$.
7. Reverse the current → force reverses (reading swaps sign).

$F = B\,I\,L$ — force is proportional to current (this practical checks that).

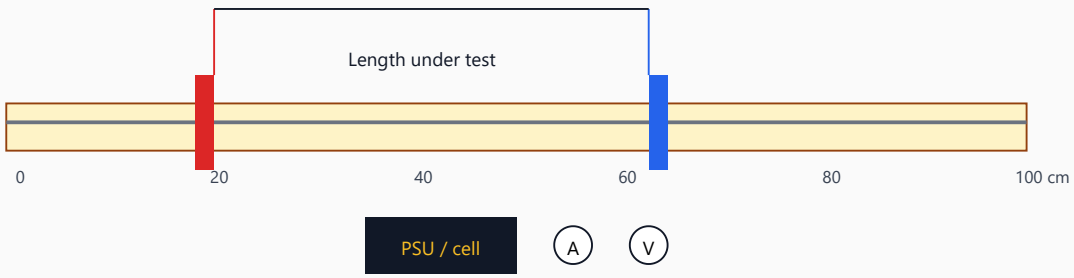
⚠️ **Safety:** Keep current values low (≤ 5 A) and limit duration — wires get hot. Use a thick wire to reduce resistance.

PAG8 — Investigating Resistance (Wire / Thermistor / LDR)

Aim: investigate how the resistance of a wire depends on its length, and how thermistor / LDR resistance varies with environment.

Wire length

Resistance wire along metre ruler, crocodile clips, circuit



Method: 1. Attach one crocodile clip to 0 cm; the second clip moves along to select a length L.
2. At each length, close the switch briefly, record V and I. *Open the switch between readings* — otherwise the wire heats up and R changes.
3. Calculate R = V / I. Plot R against L — straight line through origin. Gradient gives resistance per metre.

Thermistor (vs temperature)

Method: Immerse the thermistor in a water bath. Heat slowly. Record R (from V / I) at each temperature. Plot R against T — curve *decreases* rapidly as temperature rises (negative temperature coefficient).

LDR (vs light intensity)

Method: Place the LDR a measured distance from a lamp. Vary the distance (or light intensity using filters). Record R at each intensity. R decreases as light level rises.

⚠️ **Safety:** Keep current low (< 0.5 A for nichrome wire). Do not leave the circuit on continuously — the wire gets hot and can burn. Use insulated crocodile clips.

Exam Tip: Measure the **diameter** of the wire with a micrometer screw gauge in several places, take an average. A thinner wire has higher resistance for the same length.

Practical Skills — Generic Points (Assessed on Any PAG)

Skill	What it means
Accuracy	How close a reading is to the true value. Reduced by calibrated instruments and correct technique.
Precision	How small the scatter in repeated readings is. Improved by repeats and suitable apparatus resolution.
Resolution	The smallest change an instrument can measure. Ruler: 1 mm. Micrometer: 0.01 mm.
Random error	Unpredictable variation. Reduce by averaging repeated readings.

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Systematic error	Consistent offset (e.g. zero-error). Reduce by checking zero / calibration.
Anomalous result	A reading that doesn't fit the pattern — identify and repeat; do not include in averages.
Repeatability / Reproducibility	Repeatable: same person, same equipment, same result. Reproducible: different people/kit give the same result.

Exam Tip: When writing up a practical: state hypothesis → draw labelled diagram → list IV, DV, CVs → detail step-by-step method (so another student could copy it) → include a safety section → describe how data will be analysed.